SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I K.FLORINA(192521419) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: K. Florina

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

1.

CLOUD COMPUTING

DEFINITION

cloud Computing is a model for enabling network access to a shared pool of configuration

CHARACTERISTICS

- on-demand self service : users can access resource sustainability.
- Resource pooling : services are available over the internet.
- Rapid Elasticity : sound resources serve multiple users.

LIMITATIONS

- Dependence on internet
- security and privacy
- vendor lock-in
- Limited control over infrastructure.

DIFFERENCE FROM TRADITIONAL (ON-PREMISE

cloud Computing	on-premise
Internet based.	Local infrastrutre
pay as you go.	High upfront cost
Highly scalable.	Limited scalability
Managed by provider.	Managed by organization
accessible anywhere.	Limited remate access.
resource demand.	Resource fixed manually.

SERVICE MODELS PROVIDED

- IaaS (Infrastucture as service) → provides virtualized computing resource over the internet
- PaaS (platform as service) → provides platform tools for developing, testing and deploying applications.
- SaaS (software as a service) → provides ready-to-use software application over the internet.

2.

Cloud Computing
EVOLUTION

1. HARDWARE EVOLUTION

- Initially, organizations used physical servers
- Expensive, bulky and less efficient.

2. VIRTUALIZATION ERA

- Multiple virtual machines (VMs) can run to physical server.
- cost reduction and improved flexibility.
- Eg: VMware, Hyper-V.

3. INTERNET SOFTWARE EVOLUTION

- Rise of internet and web technologies
- web based app-lications.
- Eg: Gmail, Google Docs, office 365.

4. OUTCOME

- on demand self-service.
- Elastic scalability
- pay-per-use pricing.
- Focus on core business.

5. CLOUD COMPUTING ERA

- Computing services delivered over the internet.
- on-demand, scalable and pay-as-you-go.
- Eg: AWS, Azure.

3.

Role of Server Virtualization

Server Virtualization

Better Resourse utilization

Runs multiple VMS on one physical server

Isalation

Each VM runs independently withouth affecting.

Cost Reduction

Reduces hardware cost and maintenance

Disaster Recovery

VMs can be backed up and restored easily.

Server Virtualization VS Cloud Computing.

Virtulization	Cloud computing
creates virtual machines.	Delivers services over internet.
use hypervisor.	use both & networking.
Runs on one server	Runs in large data centers
improves hardware utilization	provides on demanand Services
Technology	Service model

4.

ENABLERS OF CLOUD COMPUTING

CLOUD

1. VIRTUALIZATION

→ creates multiple virtual machines.

→ key component: Hypervisor.

2. ABSTRACTION

* Hides the complexity of underlying hardware resources.

3. REST

→ Lightweight architectural style for web services

→ easy to use, scalable. use HTTP (GET, POST, PUT, DELETE).

4. SCALABILY & ELASTICITY

→ Resources can be scaled up or down based on demand.

→ Ensures efficient performance and cost optimization.

5. WEB SERVICES

→ Allows application over the internet.

→ Enables the interoperability.

6. CLOUD DEPLOYMENT MODELS

public cloud: Services over the internet, owned by provider.

XML